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Quiz 3: Logistic Regression
Graded Assignment • 30 min

Due Oct 30, 11:59 PM IST

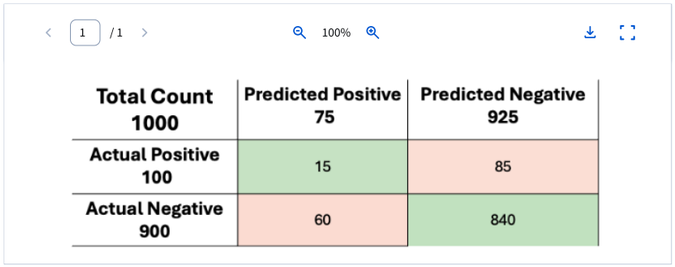
Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

Next item

2 x 2 Contingency Table

Consider the following table that summarizes the performance of a binary classifier.



Total Count 1000	Predicted Positive 75	Predicted Negative 925
Actual Positive 100	15	85
Actual Negative 900	60	840

1. 10 / 10 points

Measure	Value
Accuracy	0.855 ✓
Precision	0.2 ✓
Recall	0.15 ✓
F1	0.17 ✓

Use the preceding contingency table to determine the values for these metrics.

Response 1

- 0.2
- 0.855
- 0.15
- 0.17

Nice work

Response 2

- 0.2
- 0.15
- 0.855
- 0.17

Nice work

Response 3

- 0.17
- 0.855
- 0.15
- 0.2

Nice work

Response 4

- 0.15
- 0.17
- 0.2
- 0.855

Nice work

2. True or False: Accuracy gives a reasonable assessment of this classifiers performance. 5 / 5 points

- True
- False

Nice work

3. Consider a training example {*x*, 1} (x is in the input vector, 1 is the correct answer) in the context of **binary logistic regression**. Assume a classifier has assigned the value 0.9 to this example input during training. What is the **cross-entropy loss** for this single example? 5 / 5 points

- -log 0.9
- 0.9
- -log 0.1
- 0.1

Nice work

4. Which of the following are true of the "early stopping" approach to gradient-based training? 5 / 5 points

- [x] Early stopping requires the storage of a "checkpoint" model to revert to.

Nice work

- [x] Training can continue after an increase in the loss on the development set.

Nice work

- [x] Training continues as long as the loss on the development set declines.

Nice work

- [ ] Training continues as long as the loss on the training set declines.